QQ-F-461c
January 4, 1968
SUPERSEDING
Fed. Spec. QQ-F-461b
July 13, 1961

FEDERAL SPECIFICATION

FLOOR PLATE, STEEL, ROLLED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers raised figure, rolled steel floor plates of commercial grade for use in engine and boiler rooms, in machinery spaces, in industrial plants, in transportation equipment and in similar locations. These plates are not intended for spaces handling fuel oil or where floors with nonsparking properties are required.

1.1.1 Federal specification coverage. Federal specifications do not cover all styles, classes, and patterns which are commercially available, but only those which are generally used by the Federal Government.

1.2 Classification.

1.2.1 Styles, classes, and patterns. The floor plates shall be of the following styles, classes, and patterns, as specified (see 6.1):

Styles:

Flat back.
Hollow back.

Classes:

Class 1 - Carbon steel.
Class 2 - Copper bearing steel.

Patterns:

Pattern No. 7.
Pattern No. 8.
Pattern No. 10.
Pattern No. 12.
Pattern No. 13.
Pattern No. 14.
Pattern No. 15.
Pattern No. 16.
Pattern No. 17.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151 - Metals; Test Methods.

FSC 9515

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-163 - Steel Mill Products; Preparation for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee:

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Ill. 60606.)

American Trucking Associations, Inc.:

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P St., N. W., Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Material. Unless otherwise specified (see 6.1), floor plates shall be made of Bessemer, open-hearth, electric furnace, or basic oxygen process, carbon steel. Class 2 floor plates shall contain not less than 0.20 percent copper on ladle analysis and not less than 0.18 percent copper on check analysis.

3.2 Ingot discard. Sufficient discard shall be taken from each ingot to insure freedom from piping, undue segregation, blowholes, and other injurious defects that may result from insufficient discard.

3.3 Plate cutting. Unless planed edges are specified (see 6.1), floor plates shall have sheared or flame cut edges.

3.4 Cold bending. The floor plates shall not fracture and the web shall not separate from the raised figure when subjected to the bending test specified in 4.4.1.

3.5 Details of raised figure.

3.5.1 Design. The raised figures shall have approximately flat (or slightly rounded) tops, sloping sides, and well-defined edges where the sides and tops join. The base of the raised portions of each plate shall cover not less than 30 percent of the total surface of one side of the plate. The reverse side of the plate shall be flat, except that the portion below the raised figures on plates having a nominal weight of 2.40 and 3.00 pounds per square foot may be hollow.

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3.5.2 Patterns.

3.5.2.1 Pattern Nos. 8, 10, and 13. The floor plates with pattern numbers 8, 10, and 13 shall have raised figures, not less than 1-1/8 inches nor more than 2 inches in length at the base (see figs. 1, 2, and 3).

3.5.2.2 Pattern Nos. 7, 12, 15, and 17. The floor plates with pattern numbers 7, 12, 15 and 17 shall have raised figures not less than 5/8 inch nor more than 1-1/2 inches in length at the base (see figs. 4, 5, 6, and 7).

3.5.2.3 Pattern No. 14. The floor plates with pattern number 14 shall have raised figures not less than 5/8 inch nor more than 3/4 inch center to center, and the base of the raised figure shall be approximately square, 7/16 ± 1/16 inch on each edge (see fig. 8).

3.5.2.4 Pattern No. 16. The floor plates with pattern number 16 shall have raised figures not less than 9/16 inch nor more than 11/16 inch, center to center, and the base of the raised figures shall be approximately square, 5/16 ± 1/16 inch on each edge (see fig. 9).

3.5.3 Thickness, weight and special characteristics. The nominal thickness, weight, style, and height of raised figures shall be as specified in table I (see 6.1).

Table I. Nominal thickness and theoretical weight for rolled plate

Gage No. or thickness 1/	Theoretical weight per square foot 2/	Height of raised figure, minimum	Style
Inch	Pounds	Inch	
18	2.40	0.025	Hollow back
16	3.00	.035	Hollow back
14	3.75	.035	Flat back
13	4.50	.035	Flat back
12	5.25	.055	Flat back
1/8	6.15 3/	.055	Flat back
3/16	8.70	.060	Flat back
1/4	11.25	.060	Flat back
5/16	13.80	.060	Flat back
3/8	16.35	.060	Flat back
7/16	18.90	.065	Flat back
1/2	21.45	.065	Flat back
9/16	24.00	.065	Flat back
5/8	26.55	.065	Flat back
3/4	31.65	.065	Flat back
7/8	36.75	.065	Flat back
1	41.85	.065	Flat back

1/ Thickness is measured near edge of the plate exclusive of raised figures.

2/ Theoretical weights are shown for convenience of estimates.

3/ Pattern No 13, 6.50 pounds.

3.5.4 Shearing of plates. When specified (see 6.1), shearing shall conform to the following requirements:

3.5.4.1 Plates of pattern Nos. 14 and 16 shall be sheared parallel with the line of junction of the sides of the rib or raised figure and the web. The shearing shall be done so as to leave a full rib or full-raised figure on one edge of the plate, and, if practicable, the full width of the web between ribs or raised figures on the other edge of the plate.

3.5.4.2 Plates of pattern No. 13 shall be sheared parallel with the center line (major axis) of the raised figures and in such a manner that at least one long edge is adjacent (or tangent) to the junctions of the sides of the figures and the web.

3.5.4.3 Plates of pattern No.12 shall be sheared as near aa practical along lines at 45 degrees to the axis of the raised figures and in such a manner that at least one long edge permits the least practicable amount of web without cutting through the raised portions.

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3.6 Tolerances.

3.6.1 Thickness. Plates 1/8 inch thick and over, when ordered to thickness, shall not vary more than 0.01 inch under the ordered thickness, measured between the raised figures.

3.6.2 Weight. Floor plates 1/8 inch thick and over, when ordered to thickness, shall not exceed the nominal weight for the thickness ordered, as shown in table I, by more than the amounts permitted in table II.

Table II. Permissible overweights of plates[1/] when ordered to thickness in inches

Ordered thickness, inch	Permissible excess in average weights of lots, percent of nominal weights
1/8 to 3/16, exclusive.....	14
3/16 to 1/4, exclusive.....	10
1/4 to 5/16, exclusive.....	9
5/16 to 3/8, exclusive.....	8
3/8 to 7/16, exclusive.....	7
7/16 to 1/2, exclusive.....	6
1/2 to 5/8, exclusive.....	5.5
5/8 and over.....	5

[1/] Permissible variations in overweight for individual plates shall be 1-1/3 times the amounts given in table II. Permissible overweight for circular and sketch plates shall be 25 percent greater than the amounts given in table II. Permissible variations in overweight of a lot comprising one circular plate or one sketch plate shall be 1-2/3 times the amounts given in table II.

3.6.3 Floor plates, when ordered to weight, shall not vary from the nominal weight ordered, as shown in table I, by more than the amounts permitted in table III.

Table III. Permissible variations in weights of plates[1/] when ordered to weight

Average weight, pounds per square foot	Permissible variations in average weights of lots, percent of nominal weights	
	Over	Under
Under 3.00.....	10	5
3.00 to 6.15, exclusive.....	8	5
6.15 to 8.70, exclusive.....	6	5
8.70 to 11.25, exclusive.....	5.5	3
11.25 to 13.80, exclusive.....	5	3
13.80 to 16.35, exclusive.....	4.5	3
16.35 to 18.90, exclusive.....	4	3
18.90 to 21.45, exclusive.....	3.5	2.5
21.45 to 26.55, exclusive.....	3	2.5
26.55 to 32.40, exclusive.....	2.5	2.5

[1/] Permissible variations in weight for individual plates shall be be 1-1/3 times the amounts given in table III. Permissible overweight for circular and sketch plates shall be 25 percent greater than the amounts given in table III. Permissible variations in overweight of a lot comprising one circular plate or one sketch plate shall be 1-2/3 times the amounts given in table III.

3.6.4 Regular shearing. Unless restrictive shearing is specified (see 6.1), variations under the ordered width and length shall not exceed 1/8 inch for plates under 3/16 inch in thickness, and 1/4 inch for plates 3/16 inch in thickness, and 1/4 inch for plates 3/16 inch and over, in thickness; and variations over the ordered width and length shall not exceed the amounts permitted in table IV.

3.6.4.1 Variations in width and length may be specified all over or all under, in which case they shall be the sum of the permissible variations over and under.

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Table IV. Permissible variations over ordered width and length of regular sheared plates

Ordered width	Ordered length	Variations in inches over ordered width and length					
		Thickness under 3/8 inch		Thickness 3/8 inch to 5/8 inch, exclusive		Thickness 5/8 inch and over	
		Weight under 16.35 pounds per square foot		Weight 16.35 pounds per square foot to 26.55 pounds per square foot, exclusive		Weight 26.55 pounds per square foot and over	
		Width	Length	Width	Length	Width	Length
	Inches	Inch	Inches	Inch	Inches	Inch	Inches
To 90 inches, inclusive.	Under 120.....	3/8	1/2	7/16	5/8	1/2	1
	120 to 240, exclusive..	3/8	3/4	1/2	7/8	5/8	1-1/8
	240 to 360, exclusive..	3/8	1	1/2	1-1/8	5/8	1-1/2
	360 to 480, inclusive..	7/16	1-1/4	1/2	1-3/8	5/8	1-5/8

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3.6.5 Restrictive shearing. When restrictive shearing is specified (see 6.1), variations under the ordered width and length shall not exceed 1/16 inch for plates under 3/16 inch in thickness, and 1/8 inch for plates 3/16 inch and over in thickness; and variations over the ordered width and length shall not exceed the amounts permitted in table V (see 3.6.4.1).

3.6.6 Height of rib or raised figure. For plates 1/8 inch and over in thickness,greater than 120 inches in length, the height or rib or raised figure shall be not more than 0.010 inch under the minimum height specified in table I.

3.7 Plates of one thickness furnished under any one contract or order shall be of the same pattern, unless provision is specifically made in procurement document for more than one pattern.

3.8 Identification marking. When plates are shipped in lifts, the top plate shall be turned over and printed, stamped, or otherwise permanently marked in two places with the manufacturer's name or trademark, the specification number, composition, and pattern. When shipped loose, each plate shall be marked in one place, on the reverse side.

3.8.1 In addition to the marking specified in 3.8, the plates shall be marked with the thickness or weight per foot, the width, and length.

3.9 Workmanship. The plates shall be commercially flat and free from pits, rust, excessive scale, crimps, buckles, cracks, and other injurious defects that affect their appearance or that may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Inspection lot. All floor plates of the same type, pattern, width, and thickness and rolled from the same heat of steel, shall be considered a lot for purposes of sampling and inspection.

4.2.2 Sampling for cold-bending test and copper determination. Two longitudinal specimens, approximately 18 inches by 2 inches shall be cut, one from each of two plates for the test specified in 4.4.1. Drillings for copper analysis of class 2 floor plates shall be taken from the bend-test specimens.

4.3 Examination. The material shall be subjected to dimensional and visual examination and to inspection for preparation for shipment to assure that the requirements of this specification are met. Lots containing defective material shall be subject to rejection.

4.4 Tests. Tests shall be conducted in accordance with the applicable requirements of Fed. Test Method Std. No. 151, and as specified herein.

4.4.1 Cold-bending test. The specimen shall be bent cold, with the raised figures on the inside of the specimen, through an angle of 180°, flat on itself. If either of the bend test specimens fails to meet the requirements of 3.4, a retest will be permitted using four specimens, each from a different plate. Material which fails both the test and retest shall be rejected.

4.5 Copper analysis (class 2). To determine the copper content of class 2 floor plates, analysis shall be made of drillings from the bend test specimens.

4.6 Examination of preparation for delivery. The packaging, packing, and marking of materials shall be examined for compliance with the applicable requirements of section 5.

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Table V. Permissible variations over ordered width and length of restricted sheared plates

Ordered width	Ordered length	Variations in inches over ordered width and length					
		Thickness under 3/8 inch		Thickness 3/8 inch to 5/8 inch, exclusive		Thickness 5/8 inch and over	
		Weight under 16.35 pounds per square foot		Weights 16.35 pounds per square foot to 26.55 pounds per square foot, exclusive		Weight 26.55 pounds per square foot and over	
		Width	Length	Width	Length	Width	Length
	Inches	Inch	Inch	Inch	Inch	Inch	Inch
To 90 inches, inclusive.	Under 120	1/8	1/8	1/8	3/16	1/4	5/16
	120 to 240, exclusive	1/8	3/16	3/16	1/4	5/16	3/8
	240 to 360, exclusive	1/8	1/4	3/16	5/16	5/16	9/16
	360 to 480, inclusive	3/16	5/16	1/4	3/8	3/8	11/16

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5. PREPARATION FOR DELIVERY

(Preparation for delivery requirements specified herein apply only to direct Government procurement. Preparation for delivery requirements between contractors and sub-contractors shall be as specified in the individual order.)

5.1 Preservation and packaging.

5.1.1 Level A. Plates shall be preserved and packaged in accordance with MIL-STD-163.

5.1.2 Level C. Plates shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing.

5.2.1 Level A. Plates shall be packed in accordance with MIL-STD-163.

5.2.2 Level C. Plates, preserved and packaged by level A or C, as specified (see 6.1), shall be packed in a manner which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the Uniform Freight Classification Rules, National Motor Freight Classification Rules or other regulations as applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, the shipping containers shall be marked in accordance with MIL-STD-163.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in the procurement documents:

(a) Title, number, and date of this specification.
(b) Class, type, and pattern number (if a particular pattern is required)(see 3.1).
(c) If floor plates shall be manufactured other than specified (see 3.1).
(d) If planed edges are required (see 3.3).
(e) Size, thickness, and weight, and number of pieces (see applicable table).
(f) If regular shearing is required (see 3.5.4).
(g) If restrictive shearing is required (see 3.6.4 and 3.6.5).
(h) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

NOTE. A particular pattern should not be specified unless it is necessary for matching purposes, or of such weight as is necessary to meet engineering design requirements to the exclusion of other weights.

MILITARY CUSTODIANS:

Army - MR
Navy - SH

Review activities:

Army -MR
Navy - SH, YD

Preparing activity:

Navy - SH

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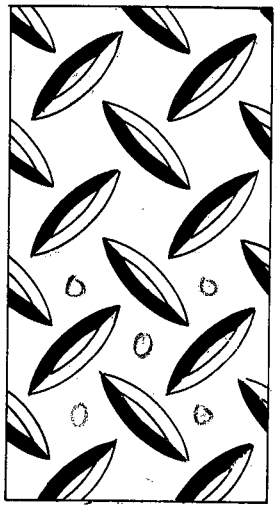


FIGURE 1. Pattern No. 8.

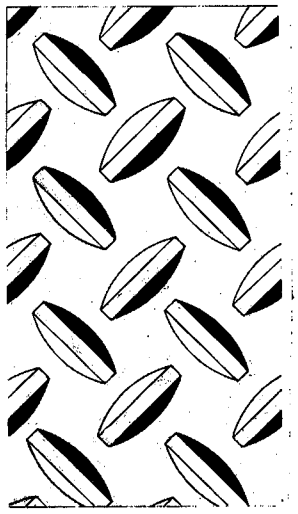


FIGURE 2. Pattern No. 10.

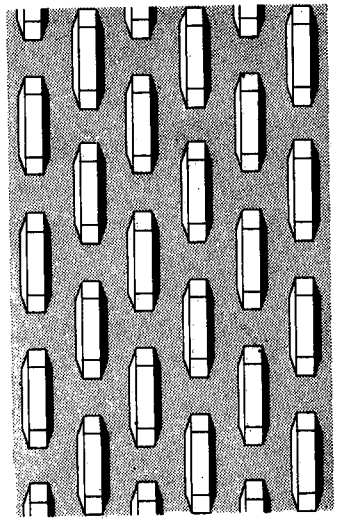


FIGURE 3. Pattern No. 13.

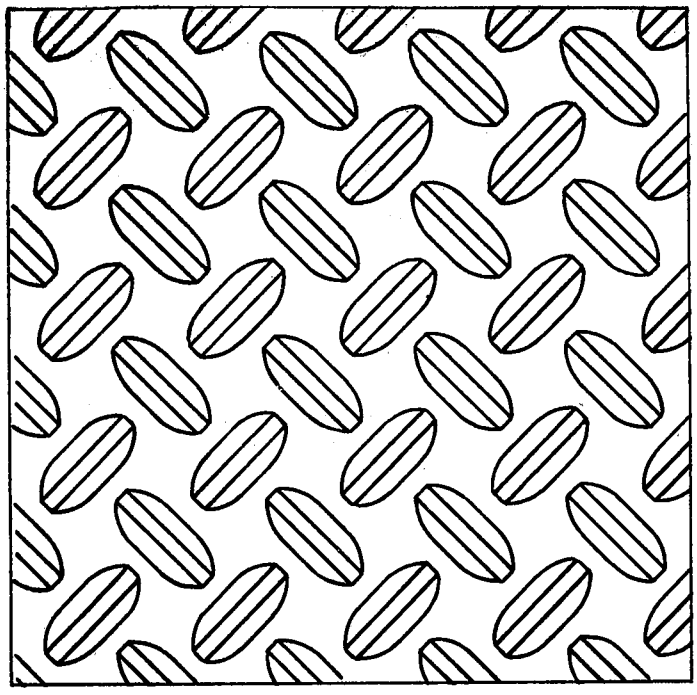


Figure 4. Pattern No. 15

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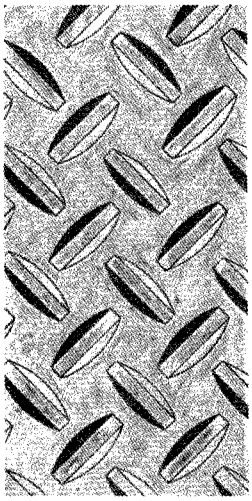


FIGURE 5. Pattern No. 7.

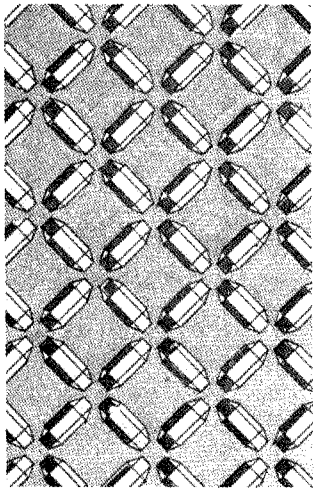


FIGURE 6. Pattern No. 12.

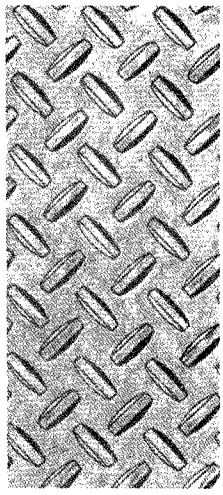


FIGURE 7. Pattern No. 17.

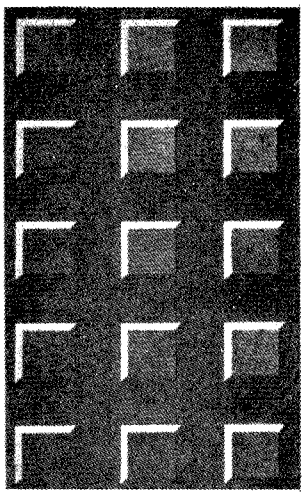


FIGURE 8. Pattern No. 14.

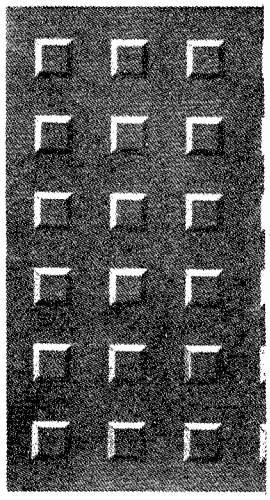


FIGURE 9. Pattern No. 16.

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain copies of documents referenced herein. Price 10 cents each.

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